

Gridlock

Predicting urban traffic bottlenecks
from road-network structure alone

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Abstract

Gridlock is an agent-based traffic simulator built on real street networks downloaded from OpenStreetMap, combined with a study of a single question: can you tell which roads in a city will congest *without simulating them*, using nothing but the shape of the network? Roads are discretised into car-length cells and cars are advanced by Nagel-Schreckenberg cellular-automaton rules, which makes congestion a continuous emergent property rather than a hand-set flag. The congestion threshold is not chosen: it is measured at 0.096 cars per cell by locating the peak of the flow-density fundamental diagram under the same parameters used on the street network. Across 100 independent trials on a 1 km radius of central Berlin, edge betweenness centrality predicts mean simulated congestion with a Spearman rank correlation of 0.801, beating four rival structural metrics by a wide margin. The entire analysis replicates on a structurally very different network in Buenos Aires, where the correlation is 0.838 and the ranking of all five metrics is unchanged. A network-level fundamental diagram shows the same regime transition as a single road, softened by the path diversity that only a real network provides.

1. Starting from the obvious model, and why it fails

The intuitive way to simulate traffic on a road network is to represent the city as a directed graph, give every car a random origin and destination, route it along the shortest path by travel time, and move it one node per time step unless the road ahead is too busy. Busy is then defined by a jam rule: if more than some fixed number of cars, say five, are already on a road segment, everyone on it stops.

That model is easy to write and wrong in three ways that matter.

It ignores road length.

A five-car limit treats a 20 m alley and a 400 m boulevard identically. The alley is saturated at five cars; the boulevard is almost empty. Any statistic computed from such a rule is really a statistic about which segments happen to be short.

Congestion is binary.

Cars either move freely or stop dead, with nothing in between. Real traffic degrades smoothly: as density climbs, mean speed falls long before anything comes to a standstill. A binary rule cannot express the very phenomenon it is supposed to be modelling.

The threshold is arbitrary.

Five is a number someone picked. It has no grounding in traffic physics and there is nothing to calibrate it against, so every downstream result inherits an unjustified free parameter.

The rest of this report replaces that model piece by piece: the representation in Section 2, the threshold in Section 3, the visualisation in Section 4. Each replacement is chosen so that the quantity it produces means something outside the simulation.

2. The model: cellular-automaton traffic on a real street graph

Gridlock keeps the OpenStreetMap graph and the shortest-path routing, and replaces the node-level representation of a car with a cell-based one drawn from the Nagel-Schreckenberg (NS) framework (Nagel & Schreckenberg, 1992).

Every road segment is subdivided into cells of five metres, roughly one car length. A cell holds either -1, meaning empty, or a non-negative integer: the speed of the car occupying it. Each time step, four rules are applied to every segment independently.

Rule	Update	What it represents
1. Accelerate	$v \leftarrow \min(v + 1, v_{\text{max}})$	drivers want to go faster
2. Brake	$v \leftarrow \min(v, \text{gap ahead})$	drivers do not hit the car in front
3. Dawdle	$v \leftarrow v - 1$ with probability p_{slow}	drivers are imperfect

4. Move	$\text{position} \leftarrow \text{position} + v$	the car advances
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Rules 1, 2 and 4 alone produce deterministic and unrealistically smooth traffic. Rule 3 is what makes the model worth running: random braking is the seed from which spontaneous phantom jams grow, and it is the reason the flow-density curve in Section 3 has the shape it does. Throughout this report $v_{\text{max}} = 5$ cells per step and $p_{\text{slow}} = 0.5$.

Cars that reach the final cell of a segment transfer to the first cell of the next segment on their route when it is free, and wait otherwise. Cars that exhaust their route are marked arrived.

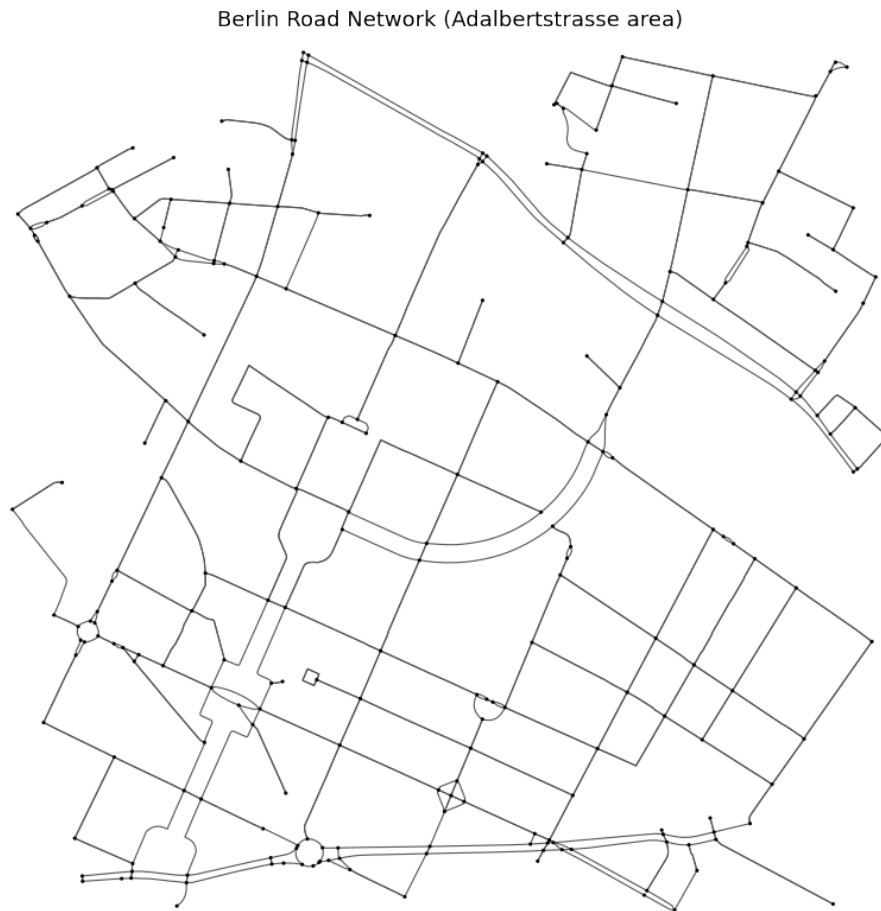


Figure 1. The 1 km radius drivable street network around Adalbertstrasse 58 in Berlin: 236 intersections and 552 road segments.

What the model captures

The simulation inherits empirically grounded road topology from OpenStreetMap, shortest-path routing weighted by travel time rather than distance, a physically motivated speed-density relationship from the NS rules, and stochastic driver behaviour through p_{slow} .

The decisive improvement over the jam-rule model is that **congestion is now continuous and emergent rather than a binary flag**. As cars accumulate on a segment, braking interactions multiply on their own and mean speed falls. That is exactly the mechanism the five-car rule was gesturing at but could not represent, because it treated every segment identically regardless of how many cells it contained.

What it leaves out, and which way that biases the results

A uniform p_{slow} across all segments is the most consequential simplification. In reality, arterial driving involves less random braking than residential streets, so the model likely overestimates congestion on fast arterials and underestimates it on slow ones.

There are no traffic signals and no intersection delays, so transfer times between segments are underestimated. In a real network this would push the effective congestion threshold slightly higher.

Uniform random origin-destination pairs overrepresent cross-city trips relative to short local ones, which biases traffic toward high-betweenness segments more than real commuting patterns would. This is worth stating plainly because it cuts in the direction of the hypothesis being tested in Section 6: it reinforces the betweenness-congestion relationship rather than threatening it, so a more realistic trip distribution would be expected to weaken the correlation, not reverse it.

Finally, transfers across parallel edges always use the first such edge, and the burn-in is 20 steps. Both are minor implementation simplifications with negligible effect on aggregate statistics.

3. Deriving the congestion threshold instead of choosing it

The arbitrary five-car rule is replaced with a threshold derived from the NS dynamics themselves. The NS model was run on an isolated 200-cell circular road across the full density range from 0 to 1, with 50 independent trials per density value and the same parameters used on the street network ($v_{\text{max}} = 5$, $p_{\text{slow}} = 0.5$). On a ring, density is a controlled input rather than an emergent outcome, which is what makes the peak locatable.

NS Model: Fundamental Diagram (flow vs. density)

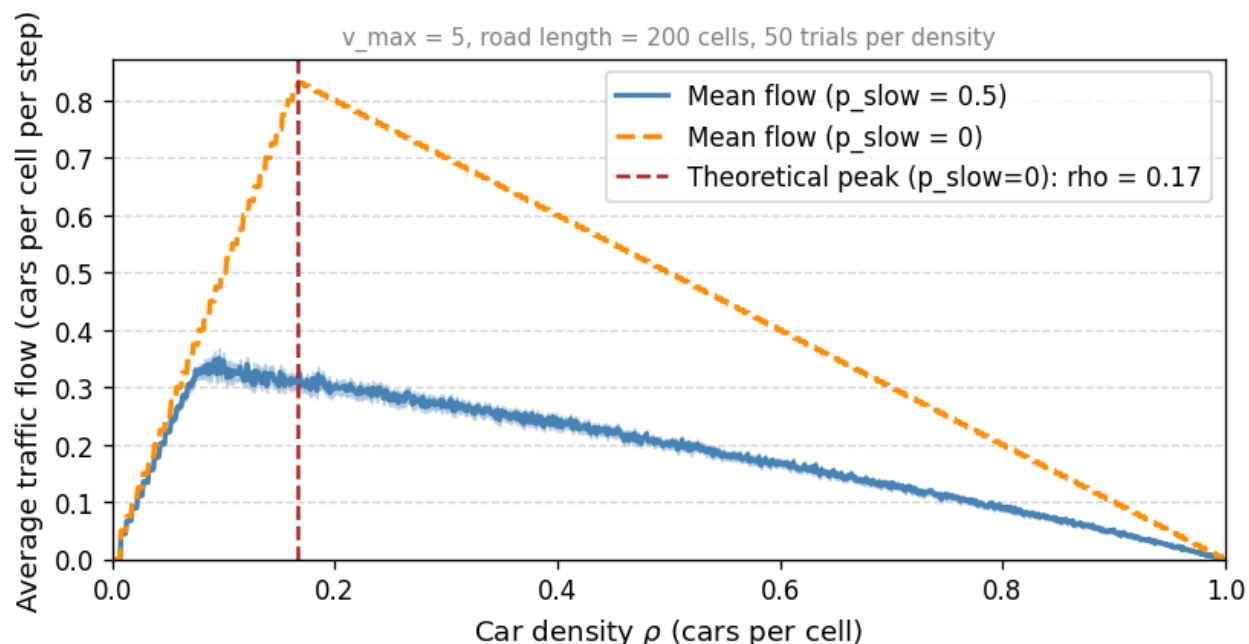


Figure 2. Flow peaks at density 0.096 for $p_{\text{slow}} = 0.5$ and at 0.168 for $p_{\text{slow}} = 0$, confirming the analytic prediction of $1/(v_{\text{max}} + 1) = 0.167$ in the zero-braking case.

The two curves do two different jobs. For $p_{\text{slow}} = 0$ the empirical peak falls at 0.168, essentially on top of the closed-form value $1/(v_{\text{max}} + 1) = 0.167$. That agreement validates the implementation against a known analytic result.

The $p_{\text{slow}} = 0.5$ curve is the one actually used. Random braking reduces mean speed at every density, compressing the flow curve and shifting its peak leftward to **density 0.096**. Above that value, adding cars reduces flow rather than increasing it, because cars are forced to brake more often than they can accelerate.

This density replaces the five-car rule. A segment's local density is

$$\text{rho}(e) = \text{cars on segment } e / \text{num_cells}(e)$$

and any segment whose $\text{rho}(e)$ exceeds 0.096 is operating in the congested regime.

Why the threshold is measured rather than derived

It is tempting to get the threshold analytically. A rough heuristic treats mean speed as falling linearly with density,

$$\text{mean speed} \approx v_{\text{max}} \times (1 - p_{\text{slow}}) \times (1 - \text{rho})$$

which makes flow

$$\text{flow}(\text{rho}) = \text{rho} \times v_{\text{max}} \times (1 - p_{\text{slow}}) \times (1 - \text{rho})$$

and setting $d(\text{flow})/d(\rho) = 0$ gives $\rho_{\text{opt}} = 0.5$.

That overestimates the true peak by roughly a factor of five. The heuristic fails because it assumes cars are spread evenly, and under NS dynamics they are not: cars **cluster**. Inside a cluster the effective gap between consecutive vehicles is far smaller than the mean gap, so braking events fire much more often than a mean-field calculation predicts. The simulated value of 0.096 accounts for the spatial structure the approximation discards, and the analytic 0.5 is best read as a loose upper bound.

Importantly, 0.096 is not a free parameter chosen by hand. It is a direct consequence of the NS rules at the specific values of v_{max} and p_{slow} in use, and it moves when they move.

4. Visualising congestion

The map is coloured by **local density** (cars divided by cells) rather than by raw car count. That normalisation is the whole point. A short segment with two cars on it may have most of its cells occupied while a long segment with two cars is almost empty; colouring by count would give them the same colour and make the map a picture of segment length.

The remaining design choices support that single number. A grey-to-gold-to-red gradient gives an intuitive traffic-light reading, segment width scales with density so hotspots survive being shrunk to thumbnail size, and a labelled colorbar in real units keeps the scale interpretable rather than decorative.

Berlin: Mean Traffic Density (single trial)
Colour and width indicate car density (cars per cell). Brighter = more congested.

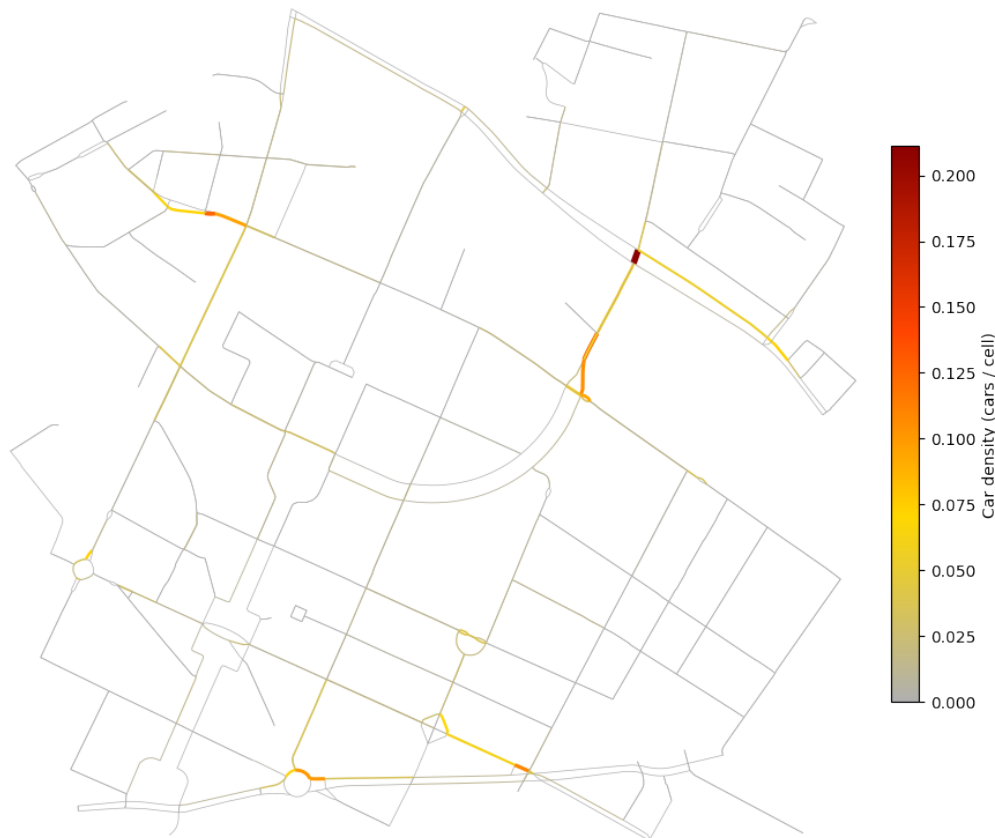


Figure 3. Even in a single trial a handful of segments carry disproportionately high density while most of the network sits near zero.

Figure 3 already shows the concentration pattern that the next two sections confirm across 100 trials: a small number of segments carry substantially more traffic than the rest, and those are precisely the segments that Section 6 identifies as having the highest betweenness centrality.

5. Which roads are consistently busy?

The empirical question is straightforward: across repeated simulations with different random origin-destination assignments, which roads consistently accumulate the most traffic?

A single trial cannot answer this. Random origin-destination pairs and probabilistic braking both inject noise, so one run conflates roads that are structurally prone to congestion with roads that happened to get unlucky draws. One hundred independent trials were therefore run with different seeds, recording mean and peak congestion per segment.

One honest limitation: 100 trials of 50 steps each, with a 20-step burn-in, leaves only 30 recorded steps per trial. The results are averages over a relatively short quasi-steady-state window, and longer runs would reduce the residual variance further. The patterns are stable enough across 100 trials for the

conclusions to hold.

Berlin: Mean Congestion Across 100 Trials
Mean car density per edge. Brighter/wider edges carry more traffic on average.

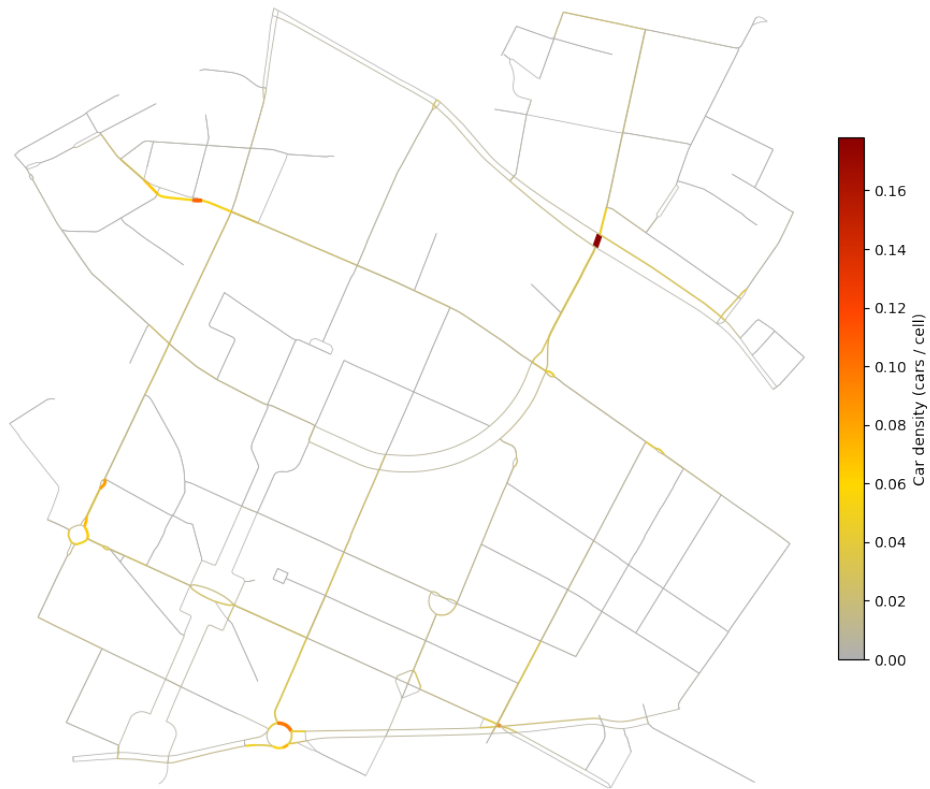


Figure 4. A small cluster of segments in the upper-centre and lower regions of the network consistently attracts the highest mean density across trials.

Berlin: Peak Congestion Across 100 Trials
Maximum car count per edge across all trials and steps.

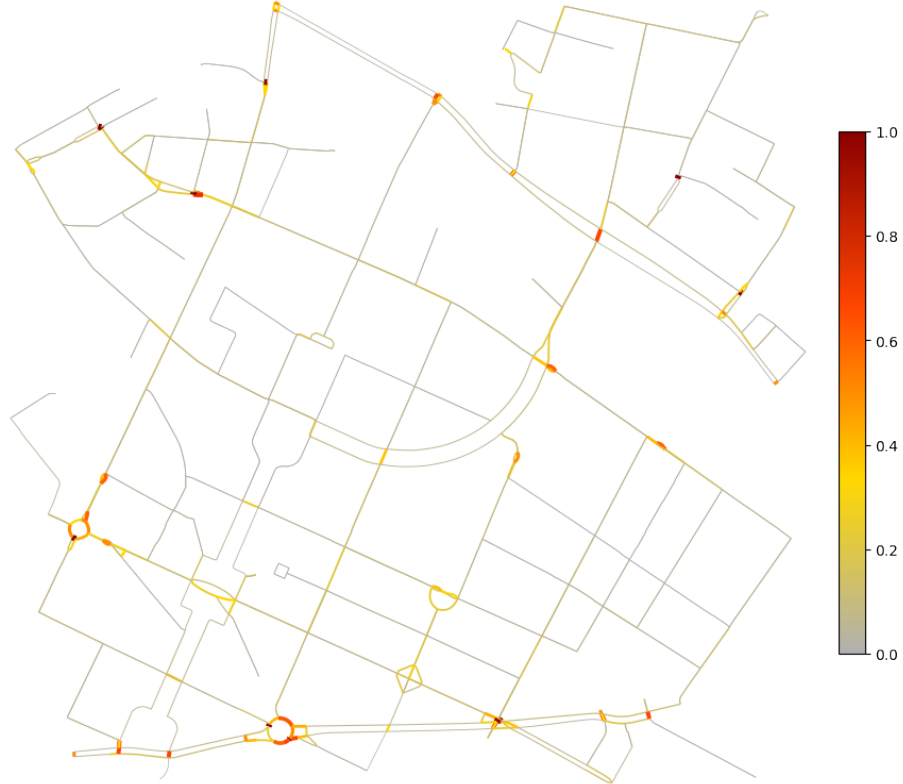


Figure 5. Peak congestion is more spatially distributed than mean congestion, with more segments reaching high density under worst-case conditions.

The two maps answer different planning questions, which is why both are kept. **Mean congestion captures chronic structural load:** which roads are busy every day. **Peak congestion captures worst-case vulnerability:** which roads are one bad afternoon away from gridlock. A road that is moderately busy every day and a road that is empty except when it seizes entirely call for different interventions.

Berlin Congestion Summary: 100 Trials

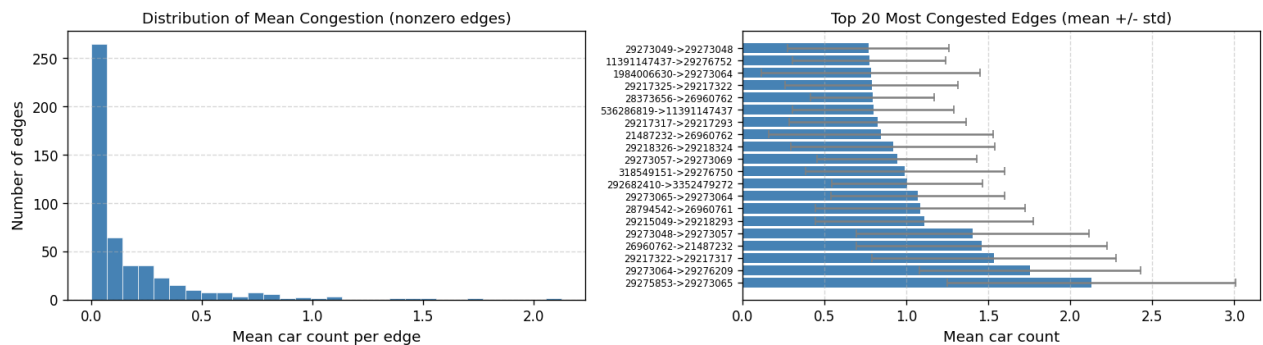


Figure 6. The distribution of mean congestion is sharply right-skewed, with most segments carrying very few cars. The busiest segment (29275853 → 29273065) averages about 2.1 cars.

The distribution across non-zero segments is heavily right-skewed: most segments carry very little traffic while a small number of high-betweenness roads carry a disproportionate share. The maximum mean congestion on any segment is 2.127 cars, with a peak of 7 cars observed at least once across all trials. This skew is also why every correlation below is measured with Spearman's rank coefficient rather than Pearson's, which the tail would otherwise dominate.

6. Predicting congestion from network structure alone

Now the question the project is really about. Running the simulation is expensive; computing a graph statistic is cheap. **Is there a structural property of the street network that predicts which roads will jam, without simulating anything?** If so, a planner could identify likely bottlenecks from a map alone, before building anything or collecting any traffic data.

The argument is short. Every car routes via the shortest path by travel time. A segment lying on many shortest paths will receive traffic from many different origin-destination pairs regardless of which specific pairs are sampled in a given trial. That is exactly what edge betweenness centrality measures. Formally, the normalised edge betweenness is

$$B(e) = [\sum_{(s,t)} \sigma(s,t,e) / \sigma(s,t)] / N(N-1)$$

where $\sigma(s,t)$ is the number of shortest paths from s to t , $\sigma(s,t,e)$ the number of those passing through segment e , and $N(N-1)$ the number of ordered node pairs in the directed graph. Since each car follows exactly one shortest path, the expected number of cars on segment e across N uniformly random origin-destination pairs is

$$E[\text{cars on } e] = N \times B(e)$$

and the expected local density is therefore

$$\text{rho}(e) = N \times B(e) / \text{num_cells}(e)$$

This yields a sharp prediction: **the segments with the highest $B(e)/\text{num_cells}(e)$ ratio are the first to enter the congested regime as N increases.** Because most segments in a city network are of broadly similar length and therefore similar cell count, betweenness centrality alone is the dominant structural predictor.

One caveat is worth stating plainly. This prediction follows from the shortest-path routing assumption, not from topology in the abstract. Under random routing, or a user-equilibrium model in which drivers react to congestion, load would spread more evenly and betweenness would be a weaker predictor. The strong correlation reported below is partly a feature of the routing model.

Five candidate metrics

Metric	Why it might predict congestion
Edge betweenness centrality	fraction of all shortest paths running through the segment

Mean endpoint degree	segments joining busy intersections may attract more routes
Mean endpoint betweenness	if a junction is on many paths, its segments should be too
Mean endpoint closeness	segments near the network's centre sit on efficient routes
Segment length	longer segments hold more cells, so perhaps more cars

Spearman rank correlation is used rather than Pearson because the congestion distribution is heavily right-skewed, as Figure 6 shows. Rank correlation is robust to that skew; Pearson would be distorted by the handful of very high congestion segments.

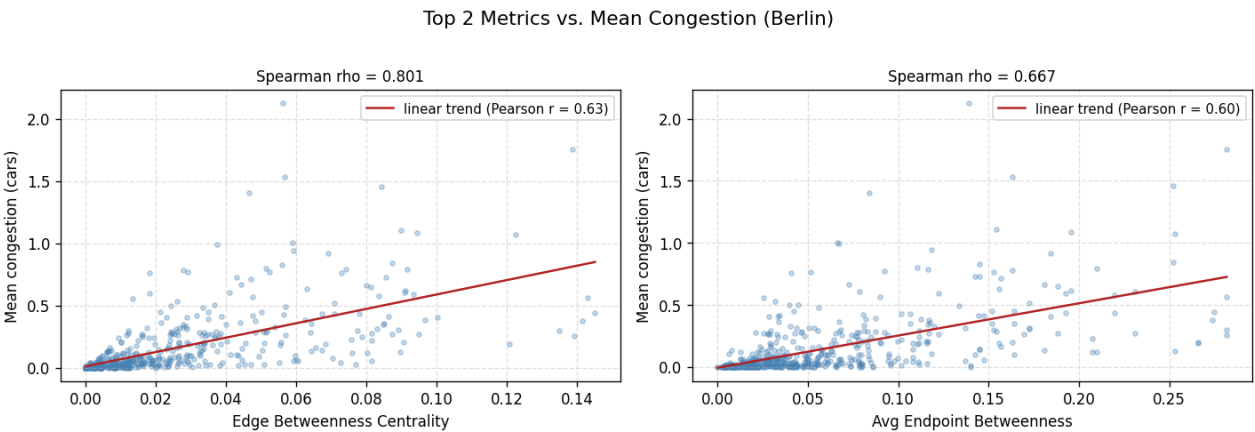


Figure 7. The positive trend is substantially tighter for edge betweenness than for mean endpoint betweenness, as the theory predicts.

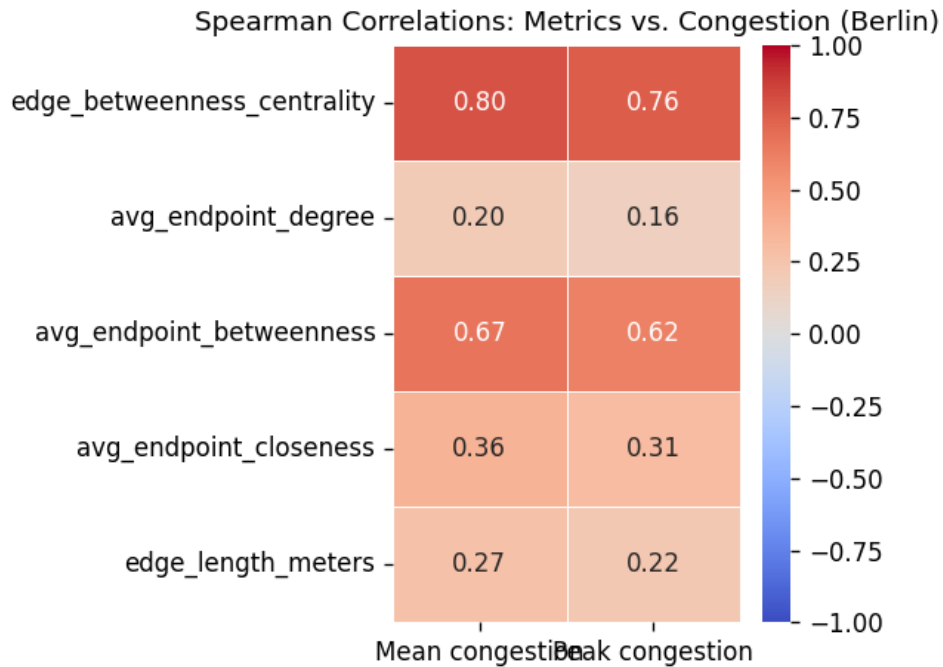


Figure 8. Edge betweenness centrality dominates every other metric by a wide margin, for both mean and peak congestion.

The results confirm the prediction.

Metric	rho (mean)	rho (peak)
Edge betweenness centrality	0.801	0.763
Mean endpoint betweenness	0.667	0.617
Mean endpoint closeness	0.364	—
Segment length	0.270	—
Mean endpoint degree	0.196	—

Edge betweenness centrality reaches Spearman $\rho = 0.801$ against mean congestion and 0.763 against peak congestion, both at $p < 0.0001$. Mean endpoint betweenness is second (0.667 and 0.617), which is exactly what the theory expects: a high-betweenness junction lies on many shortest paths, so its adjacent segments tend to as well. The node metric is a proxy for the edge metric, and it loses precision in proportion to how much local structure varies from junction to junction.

The weaker predictors are informative too. Mean endpoint closeness (0.364) measures how efficiently a junction reaches everywhere else, which correlates moderately with appearing on shortest paths but far less directly than betweenness. Segment length (0.270) is weak. Mean endpoint degree (0.196) is weakest of all, consistent with the fact that a high-degree intersection does not necessarily lie on the specific paths these cars are following: many connections is not the same as many connections that anyone actually uses.

7. Does a whole city have a fundamental diagram?

Observing that high-betweenness segments accumulate cars faster than others raises a further question: does this heterogeneous loading produce a network-level analogue of the single-road NS fundamental diagram?

To test it, the total car count was swept from 20 to 300, with 10 independent trials per condition. Mean network speed — the average speed across all occupied cells at the end of the run — serves as the network-level analogue of the NS flow measure.

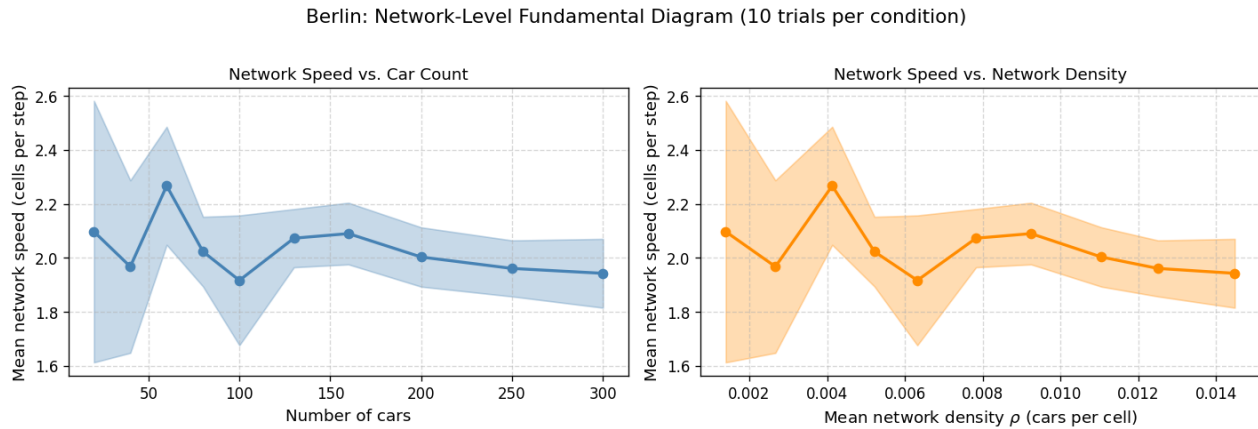


Figure 9. Speed peaks at 60 cars (2.267 cells per step) before declining, consistent with the most-loaded segments crossing $\rho_{\text{opt}} = 0.096$ first.

Speed rises from 20 to 60 cars, peaking at roughly 2.27 cells per step, then declines gradually as more cars are added. Qualitatively this matches the single-road diagram: a free-flow regime at low density and a congested regime at high density.

The transition is softer than in the one-dimensional case, and for a structural reason. **In a network, cars are not confined to a single route.** As high-betweenness segments approach the congestion threshold, some traffic redistributes onto longer alternatives, which delays the onset of system-wide congestion and smooths the speed-density curve. On a circular road a car has nowhere else to go; in a city it does, and that path diversity is exactly what blurs the regime boundary.

The peak at 60 cars is the empirical network-level optimum: the load at which average speed is highest before congestion effects begin to dominate. Beyond it, the single-segment mechanism applies at scale. The highest-betweenness segments cross $\rho_{\text{opt}} = 0.096$ first, and the braking they generate propagates backward through them, dragging down system-wide average speed.

8. Does it generalise? Validation on Buenos Aires

A result measured on one city is a result about that city. The entire analysis was therefore repeated on a deliberately different network: a 1 km radius around the Obelisco on Avenida 9 de Julio in Buenos Aires, comprising 322 intersections and 595 road segments.

This topology is structurally unlike Berlin's. It is dominated by a single very wide central avenue and several diagonal boulevards radiating outward, producing a regular radial, arterial-heavy structure, in contrast to the dense irregular grid around Adalbertstrasse. Every parameter, car count, trial count and metric was held identical.

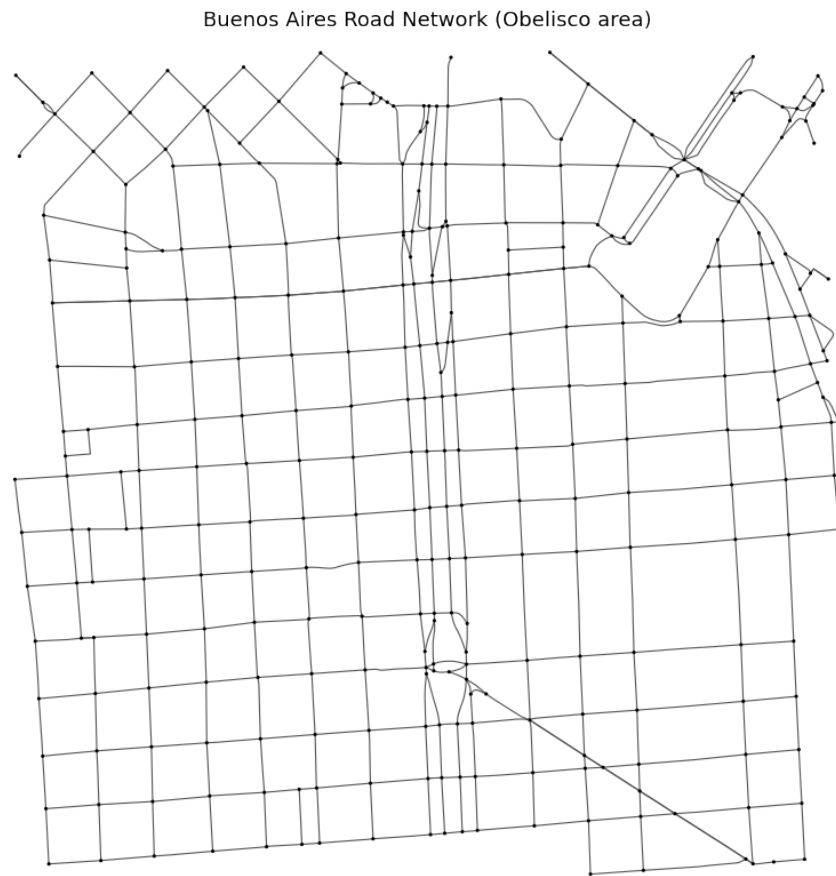


Figure 10. The network is visibly more regular than Berlin's, with Av. 9 de Julio running vertically and the diagonal boulevards creating a radial structure near the Obelisco.

Buenos Aires: Mean Congestion Across 100 Trials
Mean car density per edge. Brighter/wider edges carry more traffic on average.

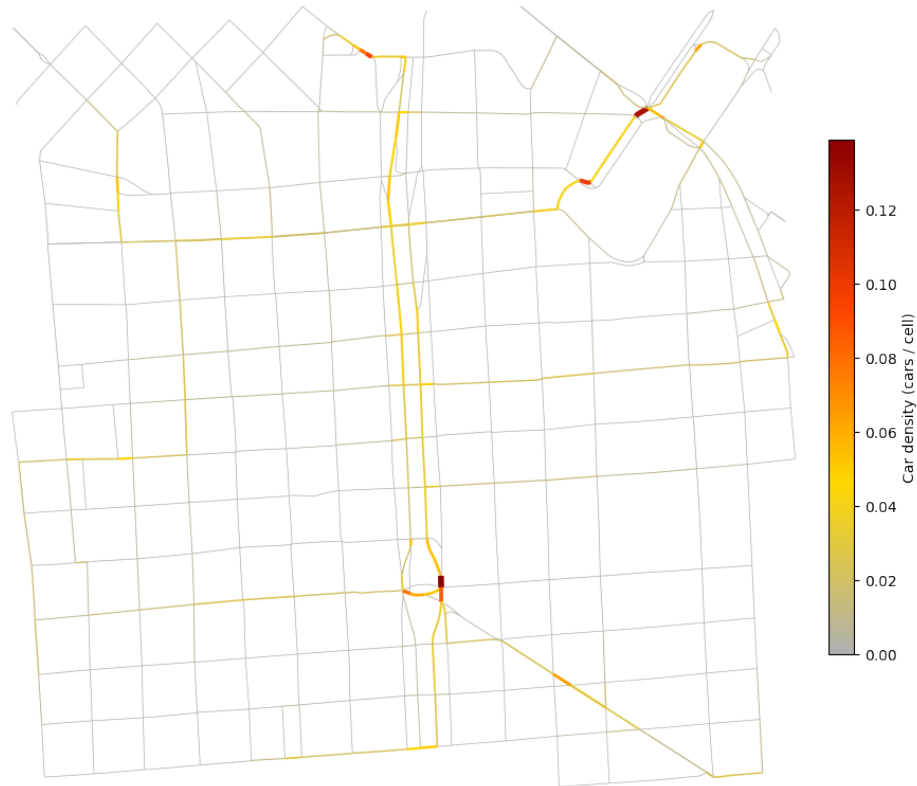


Figure 11. Congestion concentrates on the central avenue and its intersections with the diagonals, consistent with their high betweenness values.

Buenos Aires: Peak Congestion Across 100 Trials
Maximum car count per edge across all trials and steps.

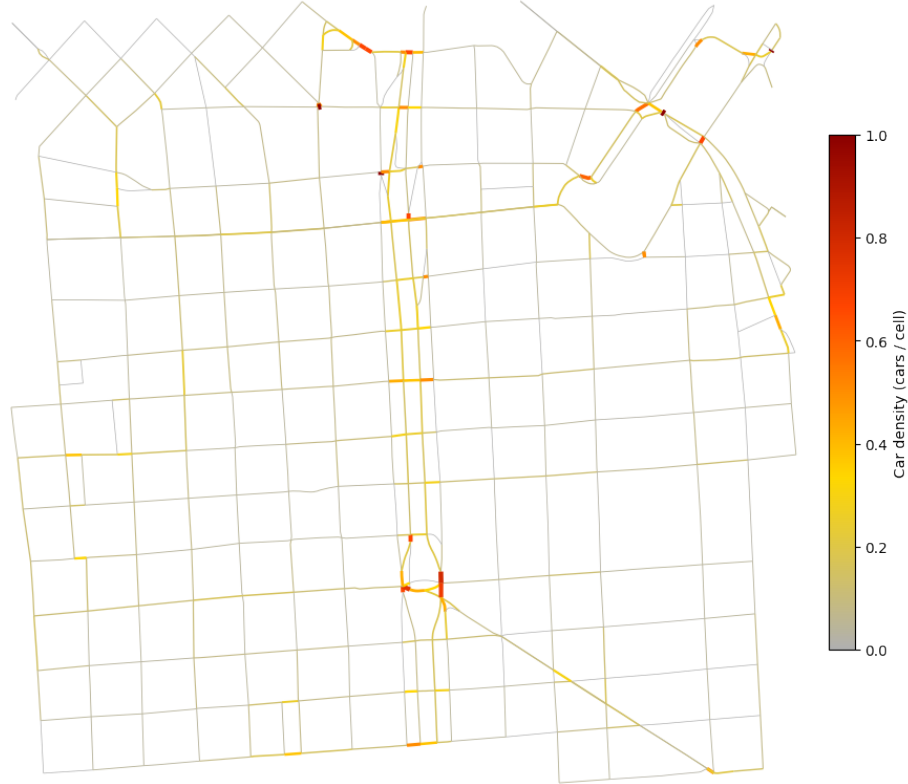


Figure 12. Peak congestion is spatially concentrated on the same arterials, indicating structural rather than random vulnerability.

Metric-Congestion Correlations: Berlin vs. Buenos Aires

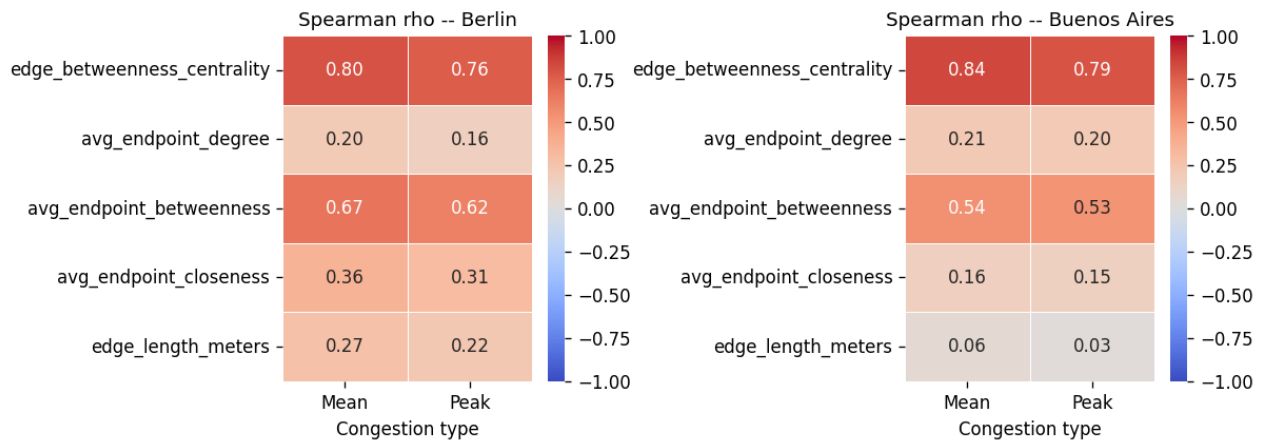


Figure 13. Edge betweenness centrality is the strongest predictor in both cities, and the ranking of all five metrics is consistent across networks.

Edge betweenness centrality reaches $\rho = 0.838$ in Buenos Aires, slightly higher than Berlin's 0.801, and remains the dominant predictor. The full metric ranking is identical across both cities, with mean

endpoint betweenness second in each case.

That consistency is the point. It suggests the predictive relationship between betweenness and congestion is not an artefact of Berlin's particular topology but a structural property of shortest-path routing that holds across quite different urban forms.

One metric does behave differently. Segment length correlates moderately with congestion in Berlin ($\rho = 0.270$) but not significantly in Buenos Aires ($\rho = 0.059$, $p = 0.15$). In the near-uniform grid around the Obelisco, segment lengths are similar across most streets and do not structurally differentiate the network. In Berlin's more irregular layout, longer segments are more likely to connect distinct neighbourhoods and therefore to appear on cross-city shortest paths. Since betweenness already captures that effect more directly, and the two are confounded, the independent contribution of segment length is likely small even in Berlin; this reading should be treated as a hypothesis rather than a finding.

9. Conclusion

Across two structurally dissimilar cities, edge betweenness centrality — computed entirely from network topology, without running any simulation — reliably identifies which roads carry the most traffic under shortest-path routing. That is the practical result: a cheap graph computation substitutes for an expensive simulation when the question is only *which roads* rather than *how badly*.

Three things make the result trustworthy rather than merely suggestive. The congestion threshold is measured from the model's own dynamics rather than chosen, so it carries no free parameter. The correlation is measured with a rank statistic appropriate to a heavily skewed distribution rather than one the tail would dominate. And the whole analysis replicates, ranking and all, on a second network chosen for being as unlike the first as a dense European inner city and a radial South American avenue system can be.

Three things bound it. The correlation is a property of shortest-path routing, and a user-equilibrium model in which drivers actively avoid congestion would weaken it. Uniform random origin-destination sampling overweights cross-city trips in a way that flatters the hypothesis. And with no traffic signals or intersection delays, the absolute congestion levels are not to be read as forecasts — it is the *ranking* of segments that the model supports, not their magnitudes.

References

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